

ITA Token CLI

A comprehensive command-line tool for managing and interacting with ITA Token on Solana. This versatile CLI provides complete control over token operations, Raydium liquidity pool management, analytics, and monitoring.

Features

Core Token Operations

Read Operations

- **Token Information** - View complete token metadata, authorities, and configuration
- **Supply Tracking** - Monitor total and circulating supply in real-time
- **Balance Queries** - Check token balances for any wallet address
- **Account Management** - Get detailed information about token accounts
- **Holder Analytics** - View top holders and distribution metrics
- **Distribution Analysis** - Analyze tokenomics and concentration metrics

Write Operations

- **Minting** - Mint new tokens to specified addresses
- **Burning** - Burn tokens to reduce supply
- **Transfers** - Send tokens between wallets
- **Batch Airdrops** - Distribute tokens to multiple addresses from CSV
- **Account Creation** - Create token accounts for wallets
- **Authority Management** - Transfer or revoke mint/freeze authorities
- **Account Closure** - Close empty accounts and reclaim rent

Raydium CLMM Integration

Pool Read Operations

- **Pool Information** - View pool reserves, fee tiers, and liquidity depth
- **Pool Discovery** - List all pools containing ITA token
- **Price Tracking** - Get real-time token price from pools
- **Position Overview** - View your active liquidity positions
- **Pool Statistics** - Monitor trading volume, fees earned, and TVL
- **Price History** - Access historical price data (24h, 7d, 30d)
- **APR Calculator** - Calculate current Annual Percentage Rate

Pool Write Operations

- **Add Liquidity** - Provide liquidity to CLMM pools
- **Remove Liquidity** - Withdraw liquidity from pools
- **Position Management** - Create, increase, decrease, or close positions
- **Fee Collection** - Claim earned trading fees from positions

- **Range Orders** - Create concentrated liquidity positions with custom price ranges

Analytics & Monitoring

- **Transaction History** - View recent token transactions
- **Volume Analytics** - Track trading volume over various timeframes
- **Market Capitalization** - Calculate current market cap
- **Tokenomics Dashboard** - Complete overview of token economics
- **Holder Growth** - Track holder count evolution
- **Whale Monitoring** - Monitor large holder activities
- **Portfolio Tracking** - View your complete ITA token portfolio value

Advanced Features

- **Holder Snapshots** - Capture holder data at specific blocks
- **Data Export** - Export holders and transactions to CSV/JSON
- **Watch Mode** - Real-time monitoring of price and volume
- **Custom Alerts** - Set notifications for price or volume thresholds
- **Multi-signature Support** - Manage tokens with multi-sig authorities



Command Reference

Configuration Management

```
# View current configuration
ita-cli config --show

# Set global configuration
ita-cli config --set cluster=mainnet --global

# Set local project configuration
ita-cli config --set rpcUrl=https://custom-rpc.com --local

# Get specific config value
ita-cli config --get cluster
```

Token Information

```
# Display complete token information
ita-cli info

# Check total supply
ita-cli supply

# Check balance of any address
ita-cli balance <address>

# View top token holders
```

```
ita-cli holders --limit 50

# Analyze token distribution
ita-cli distribution

# View tokenomics overview
ita-cli tokenomics
```

Token Operations

```
# Mint tokens
ita-cli mint --amount 1000 --to <address>

# Burn tokens
ita-cli burn --amount 500

# Transfer tokens
ita-cli transfer --to <address> --amount 100

# Batch airdrop from CSV
ita-cli airdrop --file recipients.csv

# Create token account
ita-cli create-account --owner <address>

# Set mint authority
ita-cli set-authority --type mint --new <address>

# Permanently disable minting
ita-cli disable-mint
```

Raydium Pool Operations

```
# View pool information
ita-cli pool info

# List all ITA token pools
ita-cli pool list

# Get current token price
ita-cli price

# View your liquidity positions
ita-cli pool positions

# Get pool statistics
ita-cli pool stats

# View historical price data
```

```
ita-cli price history --period 7d

# Add liquidity to pool
ita-cli pool add --amountA 1000 --amountB 5000

# Remove liquidity
ita-cli pool remove --position <positionId>

# Create new CLMM position
ita-cli pool create-position --min-price 0.9 --max-price 1.1

# Collect earned fees
ita-cli pool collect-fees --position <positionId>

# Close position
ita-cli pool close --position <positionId>
```

Analytics

```
# View recent transactions
ita-cli tx list --limit 20

# Check trading volume
ita-cli volume --period 24h

# Calculate market cap
ita-cli marketcap

# Monitor whale activities
ita-cli whales --threshold 1000000

# View your portfolio
ita-cli portfolio
```

Utility Commands

```
# Generate new wallet
ita-cli wallet generate

# Check SOL balance
ita-cli sol balance

# Request devnet SOL
ita-cli sol airdrop --amount 2

# Validate address format
ita-cli validate <address>
```

```
# Show version
ita-cli version
```

Installation

```
# Clone the repository
git clone https://github.com/yourusername/ita-token-cli.git
cd ita-token-cli

# Install dependencies
npm install

# Build the project
npm run build

# Link for global usage (optional)
npm link
```

Configuration

The CLI uses a layered configuration system with the following priority (highest to lowest):

1. **Command-line flags** - Direct overrides for single commands
2. **Environment variables** - Session-specific settings
3. **Local config** - Project-specific configuration (`.ita-cli.config.json`)
4. **Global config** - User-wide defaults (`~/.ita-cli/config.json`)
5. **Built-in defaults** - Fallback values

Configuration Options

- `cluster` - Solana cluster (devnet, testnet, mainnet-beta)
- `rpcUrl` - Custom RPC endpoint URL
- `commitment` - Transaction commitment level (processed, confirmed, finalized)
- `walletPath` - Path to wallet keypair file
- `programId` - ITA Token program ID
- `tokenMint` - ITA Token mint address

Environment Variables

```
export ITA_CLUSTER=devnet
export ITA_RPC_URL=https://api.devnet.solana.com
export ITA_WALLET=~/.config/solana/id.json
```

Example Workflows

Initial Setup

```
# Configure for devnet
ita-cli config --set cluster=devnet --global
ita-cli config --set walletPath=~/.config/solana/id.json --global

# View token information
ita-cli info
ita-cli supply
```

Managing Tokens

```
# Mint new tokens
ita-cli mint --amount 10000 --to YOUR_WALLET_ADDRESS

# Check your balance
ita-cli balance YOUR_WALLET_ADDRESS

# Transfer to another wallet
ita-cli transfer --to RECIPIENT_ADDRESS --amount 1000

# Burn excess supply
ita-cli burn --amount 500
```

Managing Liquidity

```
# Check current pools
ita-cli pool list

# View pool details
ita-cli pool info

# Add liquidity
ita-cli pool add --amountA 5000 --amountB 10000

# Monitor your positions
ita-cli pool positions

# Collect fees
ita-cli pool collect-fees
```

Monitoring & Analytics

```
# Watch price in real-time
ita-cli watch price

# Check holder distribution
```

```
ita-cli holders --limit 100
ita-cli distribution

# Export holder data
ita-cli export holders --format csv --output holders.csv

# View trading volume
ita-cli volume --period 7d
```

Security Best Practices

- **Never commit wallet keypairs** to version control
- **Use environment variables** for sensitive configuration
- **Store keypairs securely** with appropriate file permissions
- **Verify addresses** before sending large amounts
- **Test on devnet** before mainnet operations
- **Use hardware wallets** for production/mainnet operations
- **Review transactions** before confirming

Troubleshooting

Common Issues

Connection Errors

```
# Test connection
ita-cli config --get rpcUrl
# Try alternative RPC
ita-cli config --set rpcUrl=https://alternative-rpc.com
```

Insufficient SOL for Transactions

```
# Check SOL balance
ita-cli sol balance
# Request devnet airdrop
ita-cli sol airdrop --amount 2
```

Invalid Wallet Path

```
# Verify wallet exists
ls -la ~/.config/solana/id.json
# Update wallet path
ita-cli config --set walletPath=/correct/path/to/keypair.json
```

Contributing

Contributions are welcome! Please feel free to submit a Pull Request.

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Links

- [Solana Documentation](#)
- [Raydium Documentation](#)
- [SPL Token Program](#)

Support

For issues and questions:

- Open an issue on GitHub
- Join our Discord community
- Email: support@itatoken.com

⚠ Disclaimer: This tool is provided as-is. Always verify transactions and test thoroughly on devnet before using on mainnet. The developers are not responsible for any loss of funds.